Finding nth min, nth max array elements:

TC

File Edit Run Compile Project Options Debug

Line 8 Col 2 Insert Indent Tab Fill Unindent * E

```c
#include<stdio.h> #include<conio.h>
void main()
{int a[100],i,n,j,t,max, min;clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{for(j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
printf("\nEnter nth min, nth max values ");
scanf("%d%d",&min,&max);
for(i=1;i<n;i++)
{if(a[i]>a[i-1])
{min--;if(min==1){printf("\nMin=%d\n",a[i]);break;}}}
for(i=n-2;i>=0;i--)
{if(a[i]<a[i+1])
{max--;if(max==1){printf("Max=%d",a[i]);break;}}}
getch();
}
```

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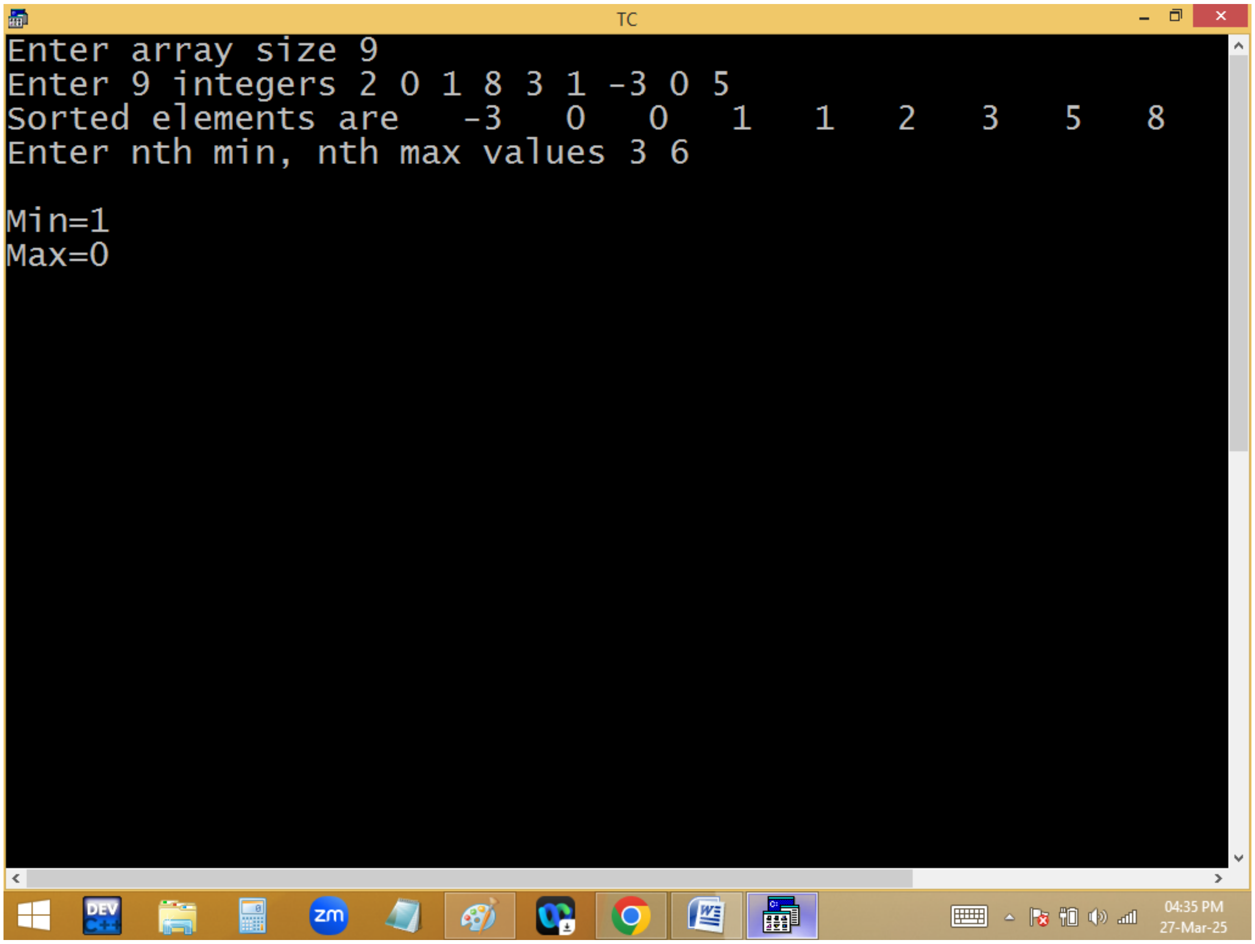
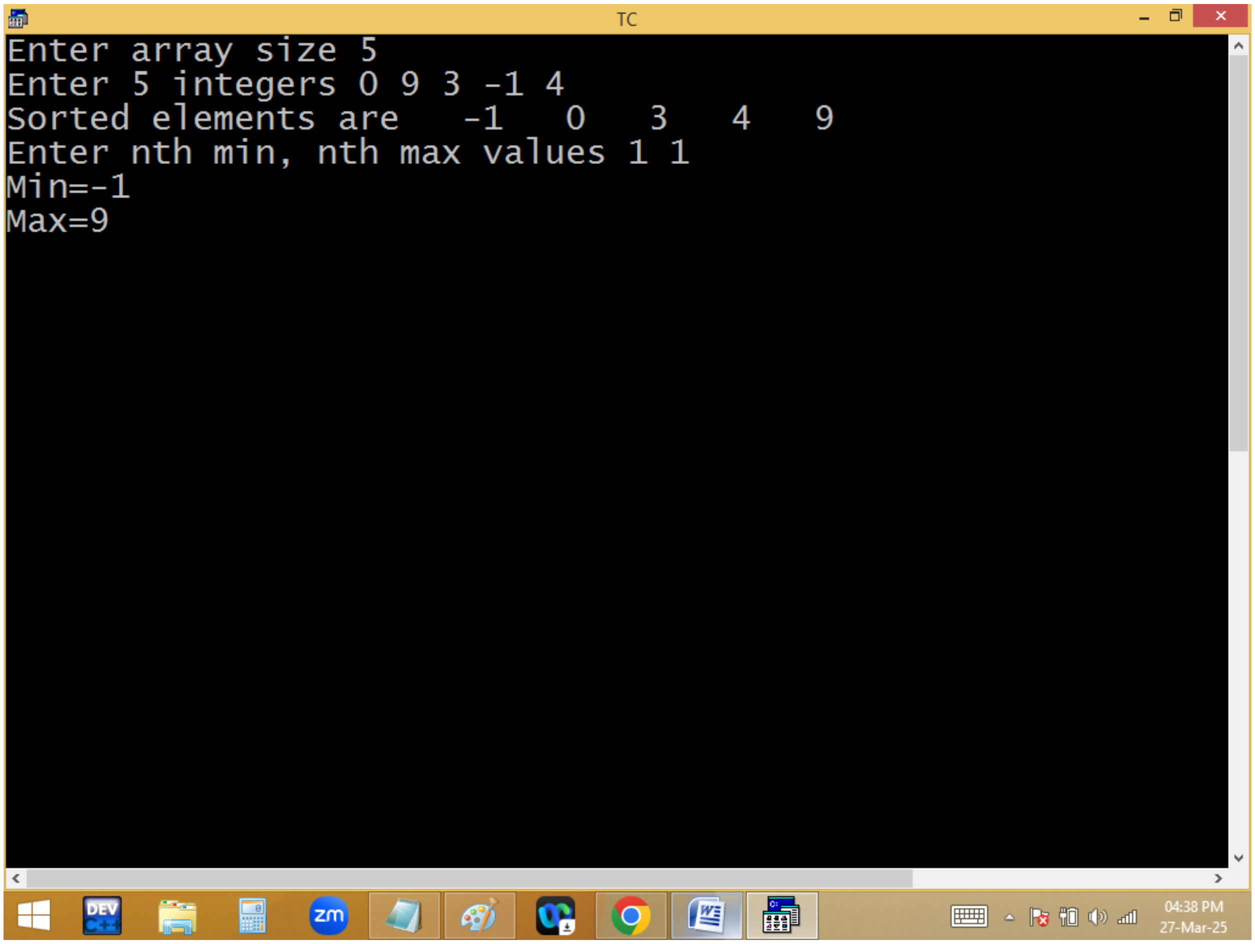



```c
#include<stdio.h>

#include<conio.h>

void main()

{

int a[100],i,n,j,t,max, min;clrscr();

printf("Enter array size ");scanf("%d",&n);

printf("Enter %d integers ",n);

for(i=0;i<n;i++)scanf("%d",&a[i]);
```

```
for(i=0; i<=n-2;i++)

{

for(j=0;j<=n-i-2;j++)

{if(a[j]>a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}

}

printf("Sorted elements are ");

for(i=0;i<n;i++)printf("%4d",a[i]);

printf("\nEnter nth min, nth max values ");

scanf("%d%d",&min,&max);

if(min==1)printf("Min=%d\n",a[0]);

else

{

for(i=1;i<n;i++)

{

if(a[i]>a[i-1])

{

min--;

if(min==1)
```

```
{printf("\nMin=%d\n",a[i]);break;}
}
}
}
if(max==1)printf("Max=%d",a[n-1]);
else
 {
for(i=n-2;i>=0;i--)
{
if(a[i]<a[i+1])
{
max--;
if(max==1)
{
printf("Max=%d",a[i]);break;
}
}
}
```

```
}
getch();
}
```


TC
Enter array size 5
Enter 5 integers 0 9 3 -1 4
Sorted elements are   -1   0   3   4   9
Enter nth min, nth max values 1 1
Min=-1
Max=9
DEV
zm
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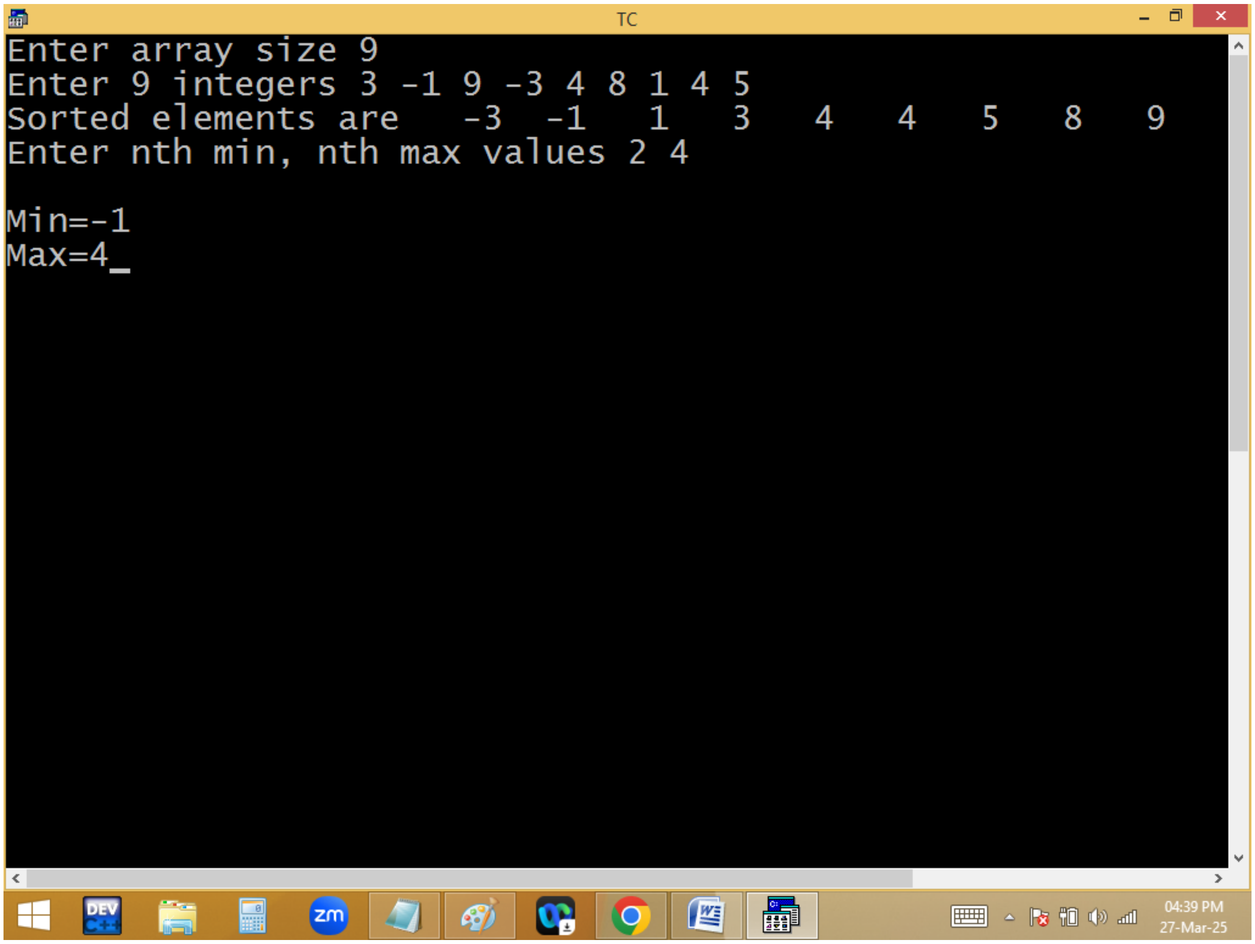
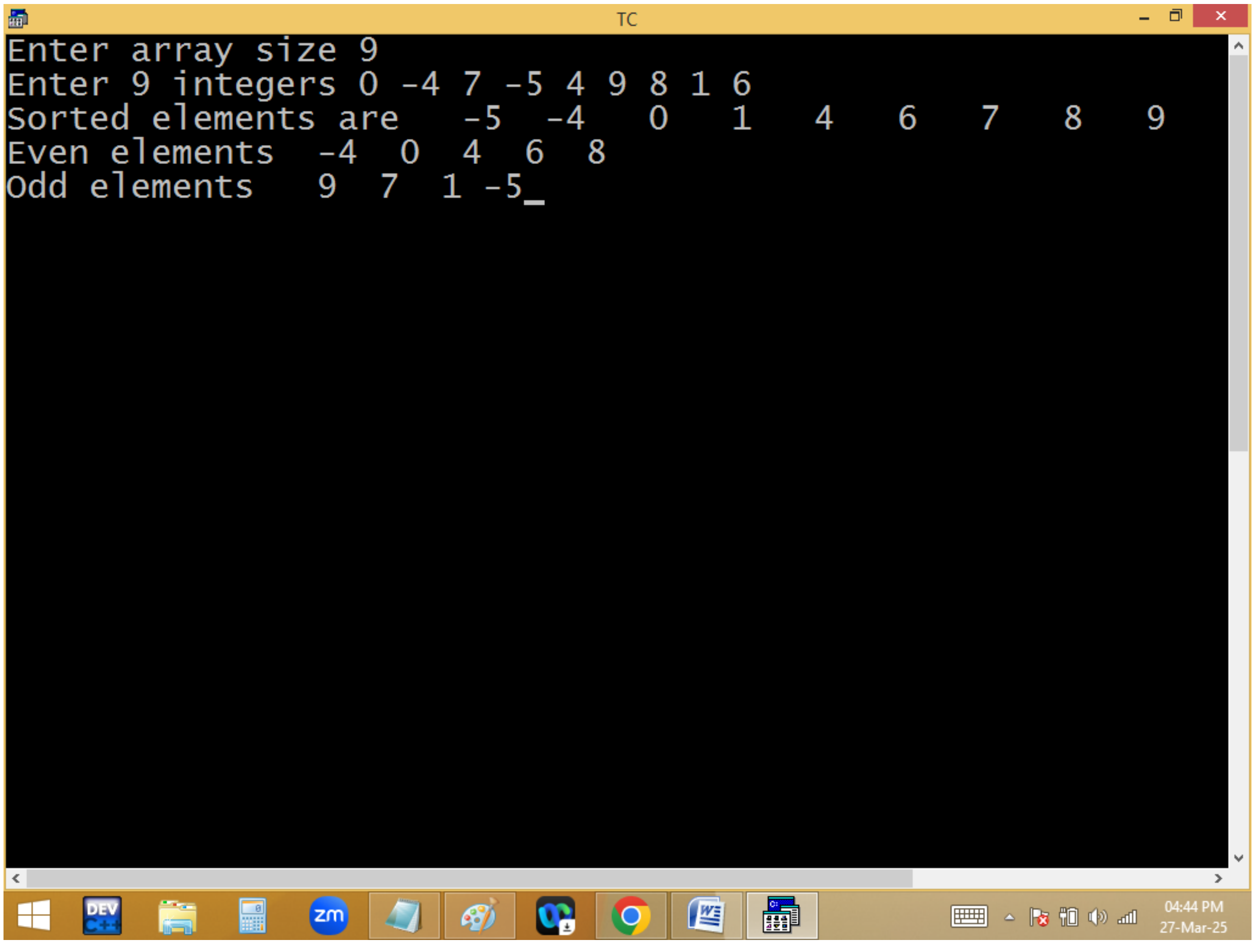
27-Mar-25

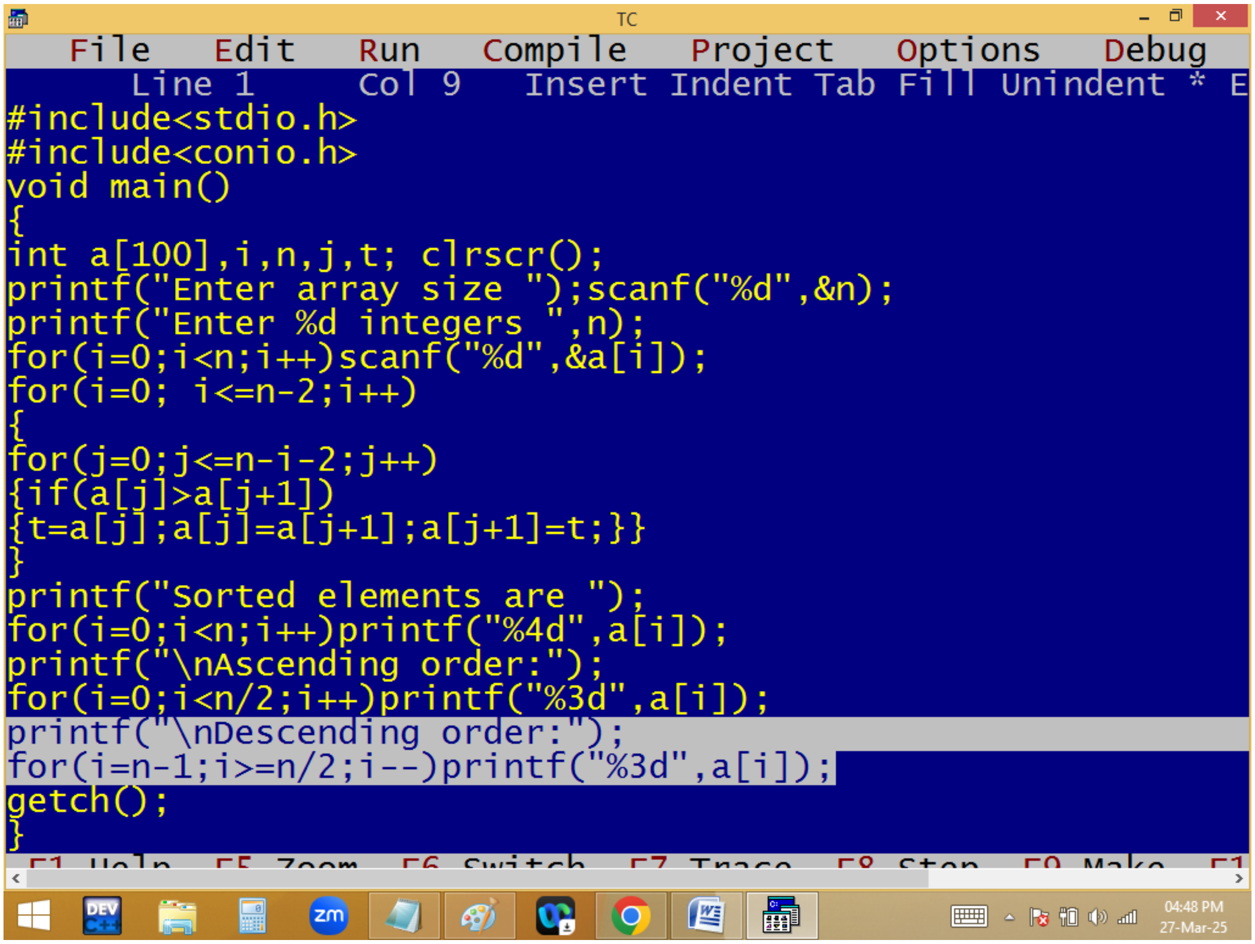


Even elements in ascending and odd elements in descending:

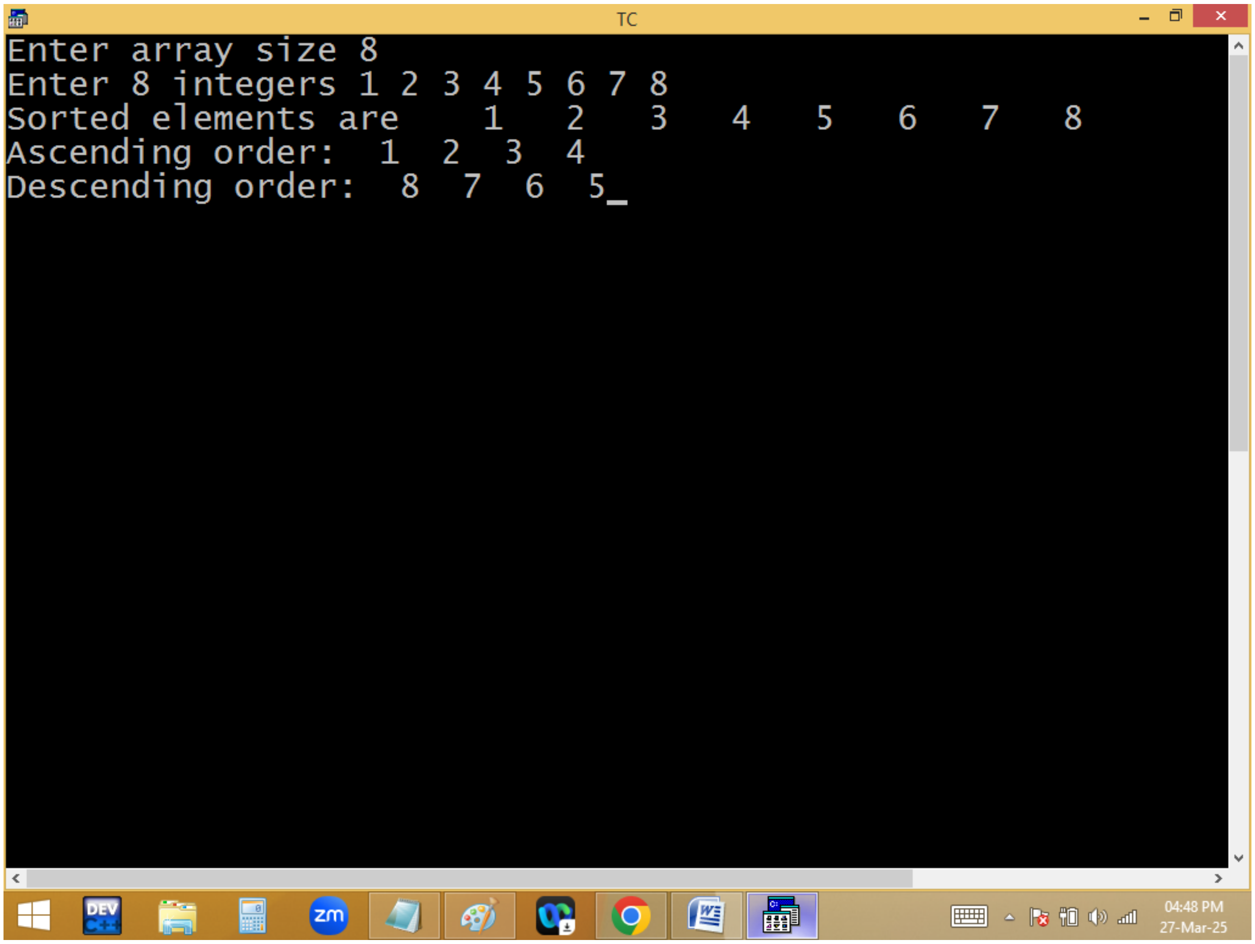
```
File    Edit    Run    Compile    Project    Options    Debug
     Line 13     Col 31   Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t; clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1])
{t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
printf("\nEven elements ");
for(i=0;i<n;i++)if(a[i]%2==0)printf("%3d",a[i]);
printf("\nOdd elements ");
for(i=n-1;i>=0;i--)if(a[i]%2!=0)printf("%3d",a[i]);
getch();
}
```



Half elements in ascending , half elements descending order:




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t; clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1])
{t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
printf("\nAscending order:");
for(i=0;i<n/2;i++)printf("%3d",a[i]);
printf("\nDescending order:");
for(i=n-1;i>=n/2;i--)printf("%3d",a[i]);
getch();
}
```


TC
Enter array size 8
Enter 8 integers 1 2 3 4 5 6 7 8
Sorted elements are   1   2   3   4   5   6   7   8
Ascending order: 1 2 3 4
Descending order: 8 7 6 5
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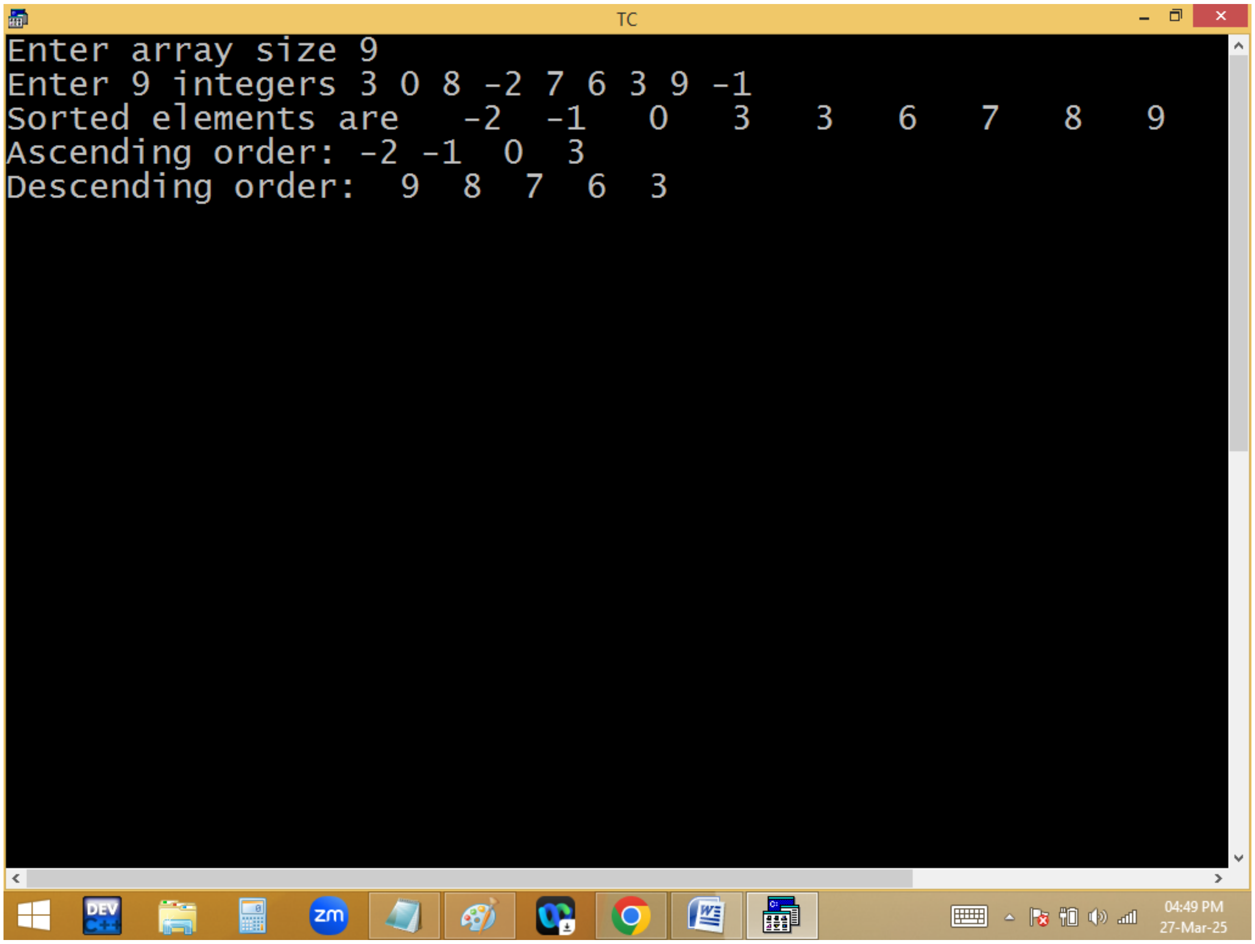
27-Mar-25

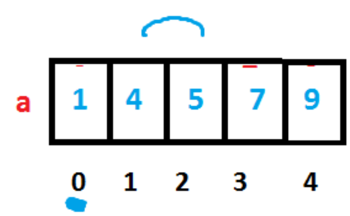



```
for( i=1;i<5;i++ )
{
if( a[i]>a[i-1]) min--;
if(min==1) p(min=a[i]);
break;
}
```

5

```
for( i=n-2; i>=0;i--)
{
if( a[i]<a[i+1])max--;
if(max==1)p(a[i]);break;
}
```

4

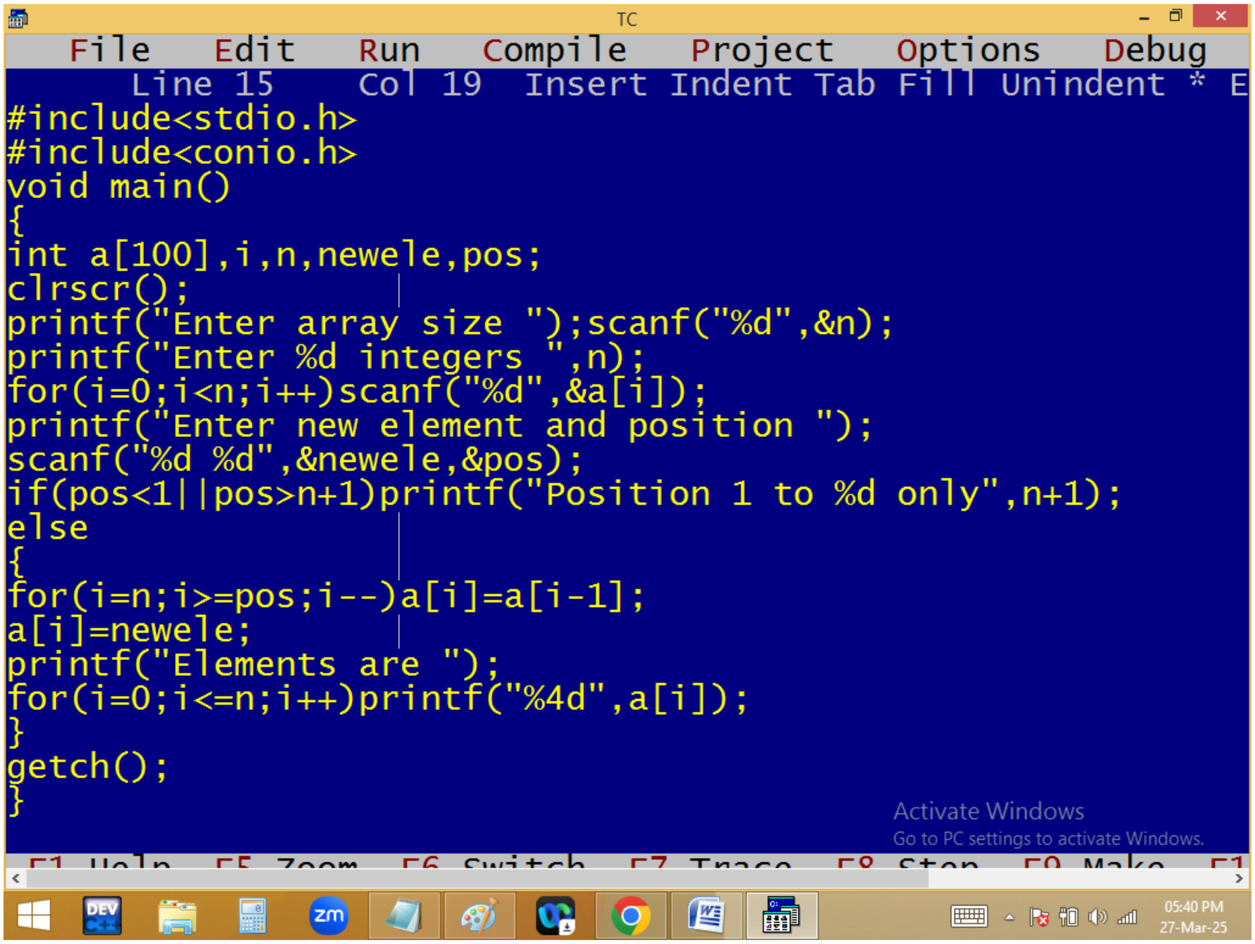


i	i-1
1	0
2	

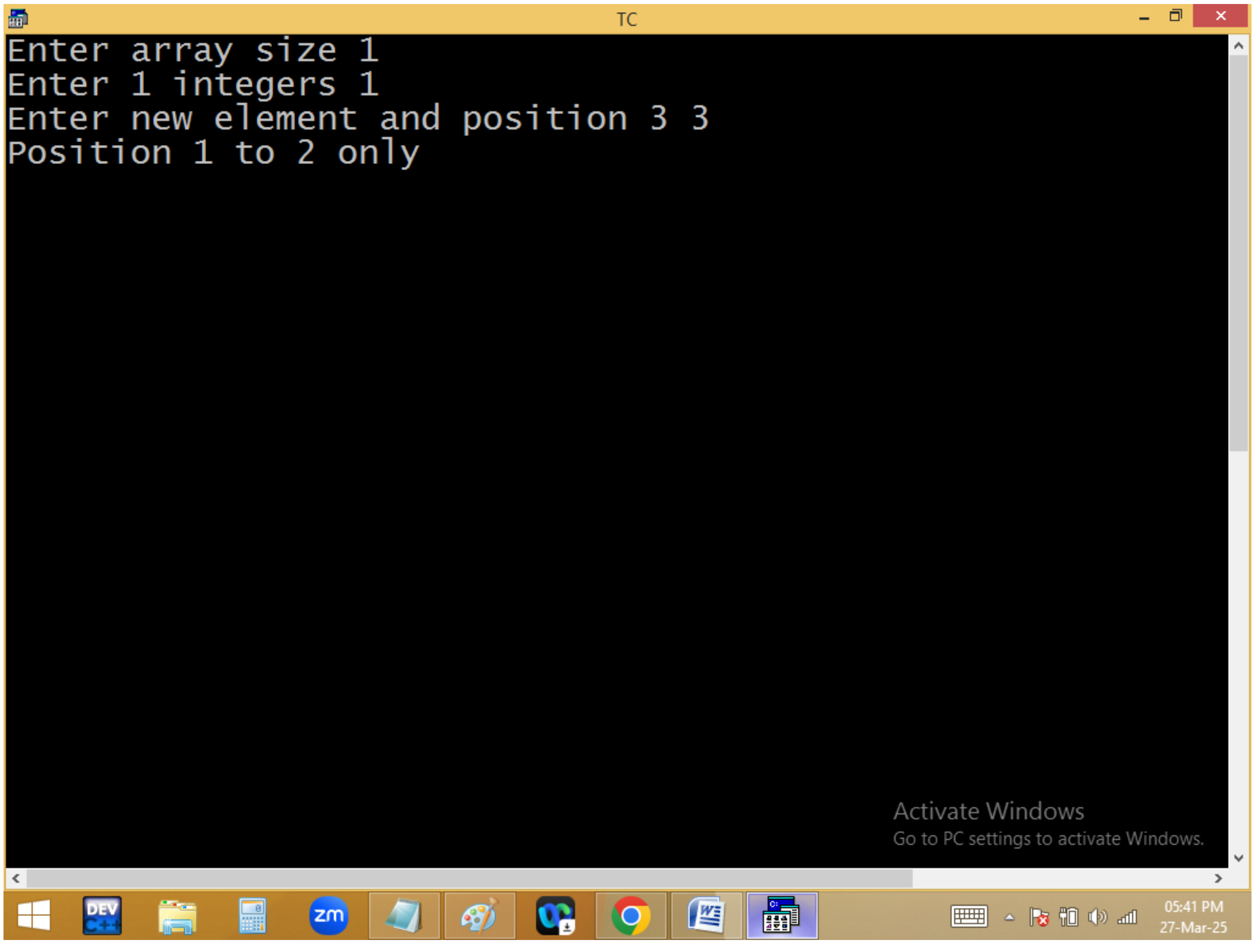
3rd min
~~3~~
~~2~~
1

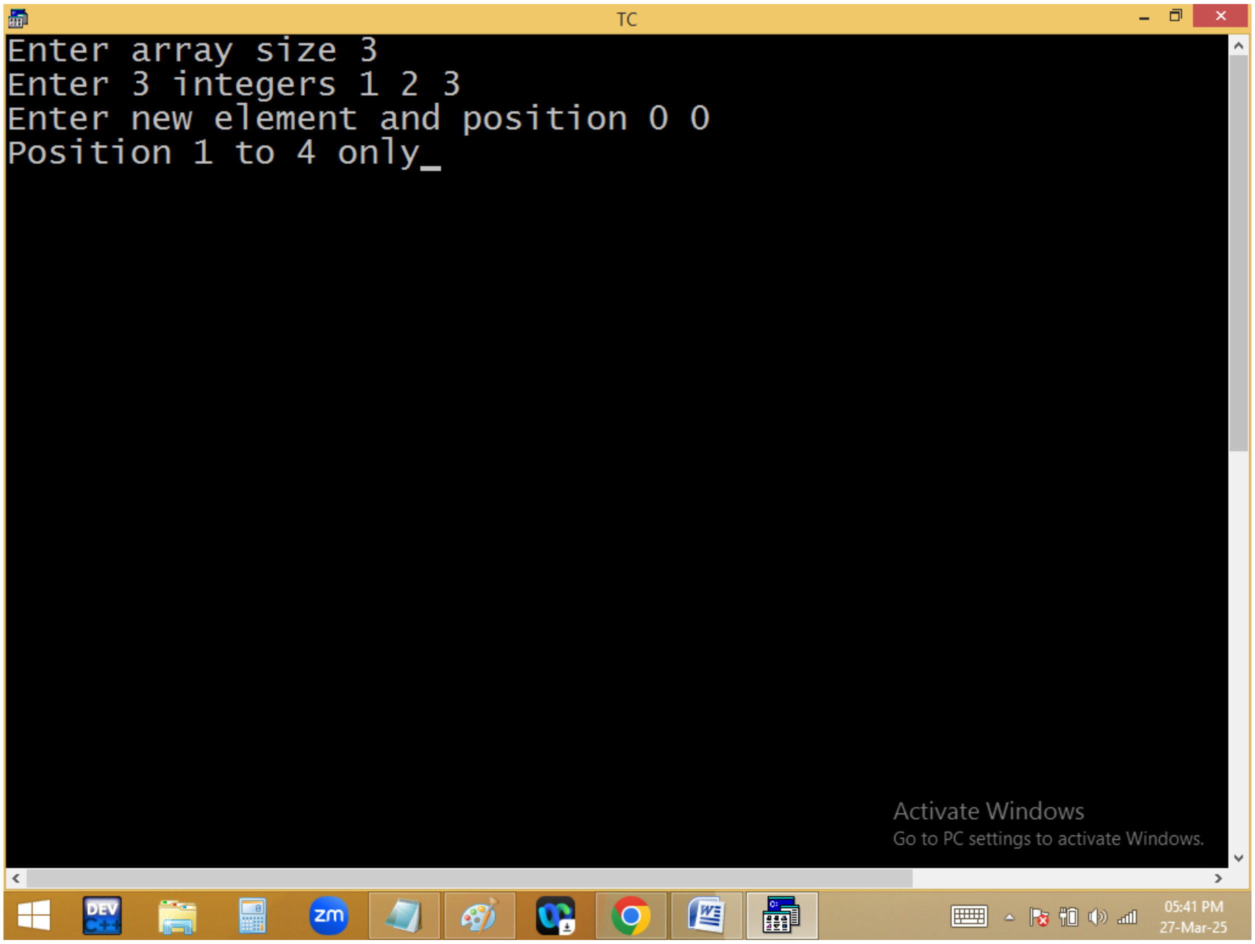
4th max	i	i+1
~~4~~	3	4
~~3~~	2	3
~~2~~	1	2
1		

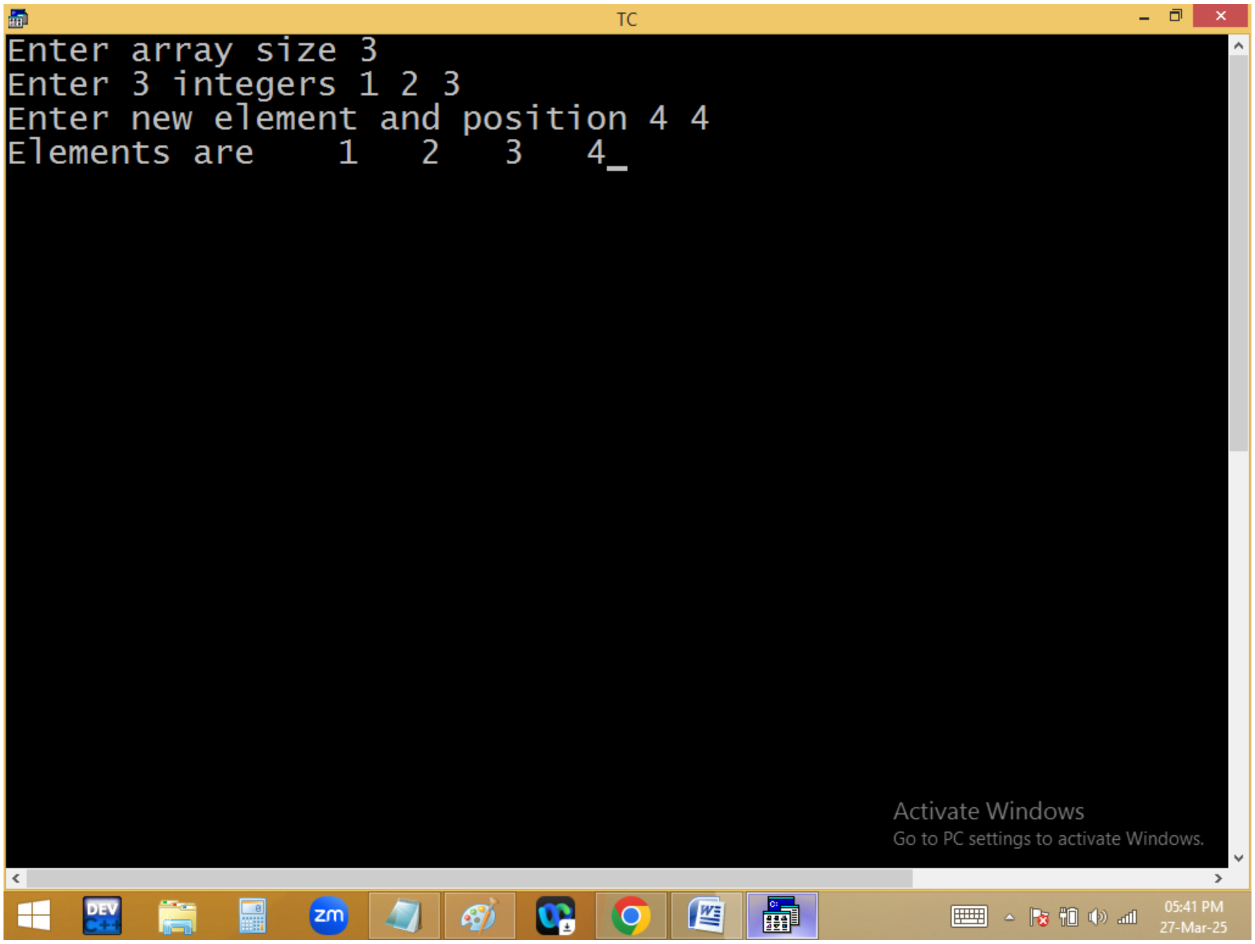
Inserting a new element in a particular position of array, without damaging existing element?




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,newele,pos;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter new element and position ");
scanf("%d %d",&newele,&pos);
if(pos<1||pos>n+1)printf("Position 1 to %d only",n+1);
else
{
for(i=n;i>=pos;i--)a[i]=a[i-1];
a[i]=newele;
printf("Elements are ");
for(i=0;i<=n;i++)printf("%4d",a[i]);
}
getch();
}
```


TC
Enter array size 1
Enter 1 integers 1
Enter new element and position 3 3
Position 1 to 2 only
Activate Windows
Go to PC settings to activate Windows.
05:41 PM
27-Mar-25


TC
Enter array size 3
Enter 3 integers 1 2 3
Enter new element and position 0 0
Position 1 to 4 only_
Activate Windows
Go to PC settings to activate Windows.
05:41 PM
27-Mar-25


TC
Enter array size 3
Enter 3 integers 1 2 3
Enter new element and position 4 4
Elements are 1 2 3 4
Activate Windows
Go to PC settings to activate Windows.
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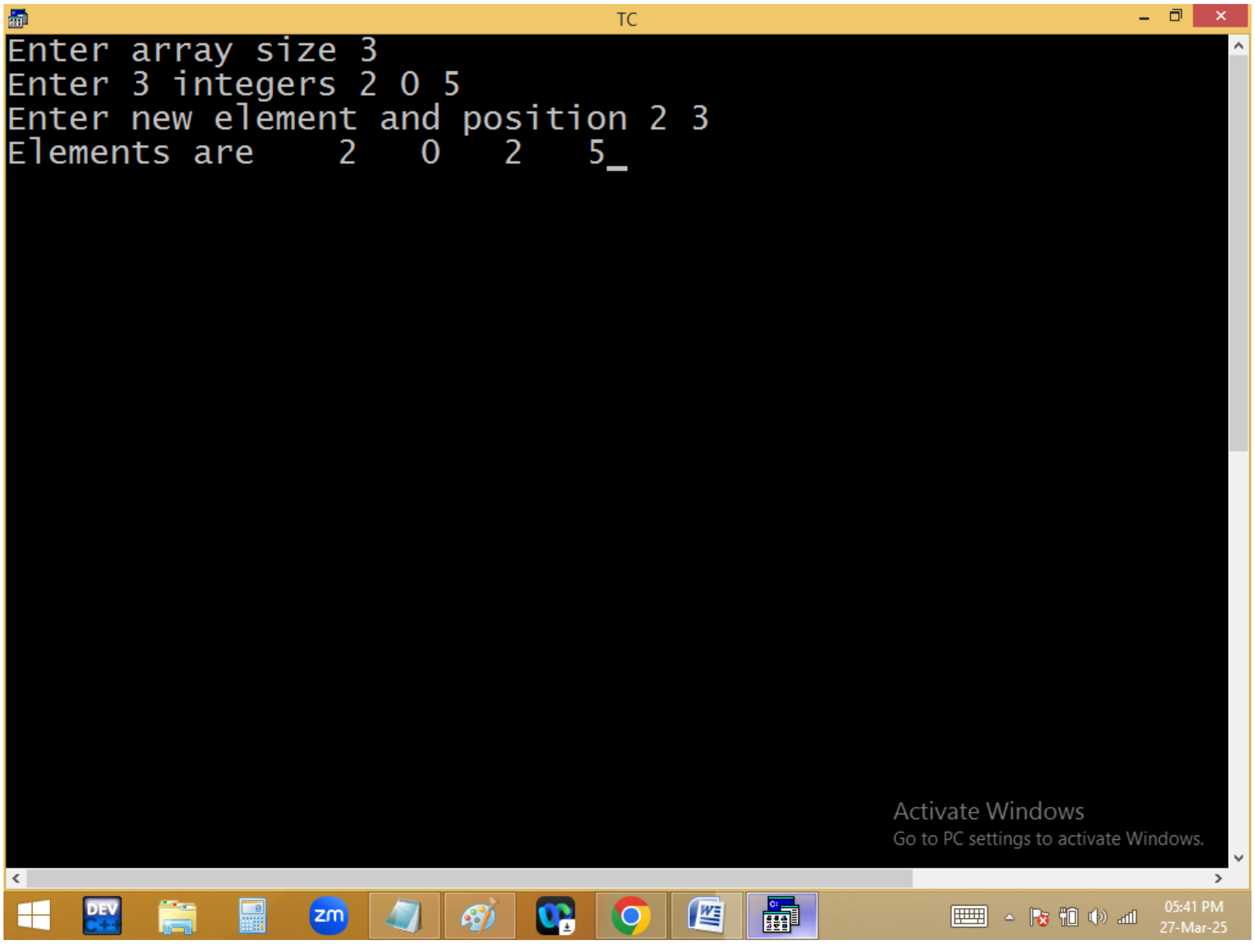
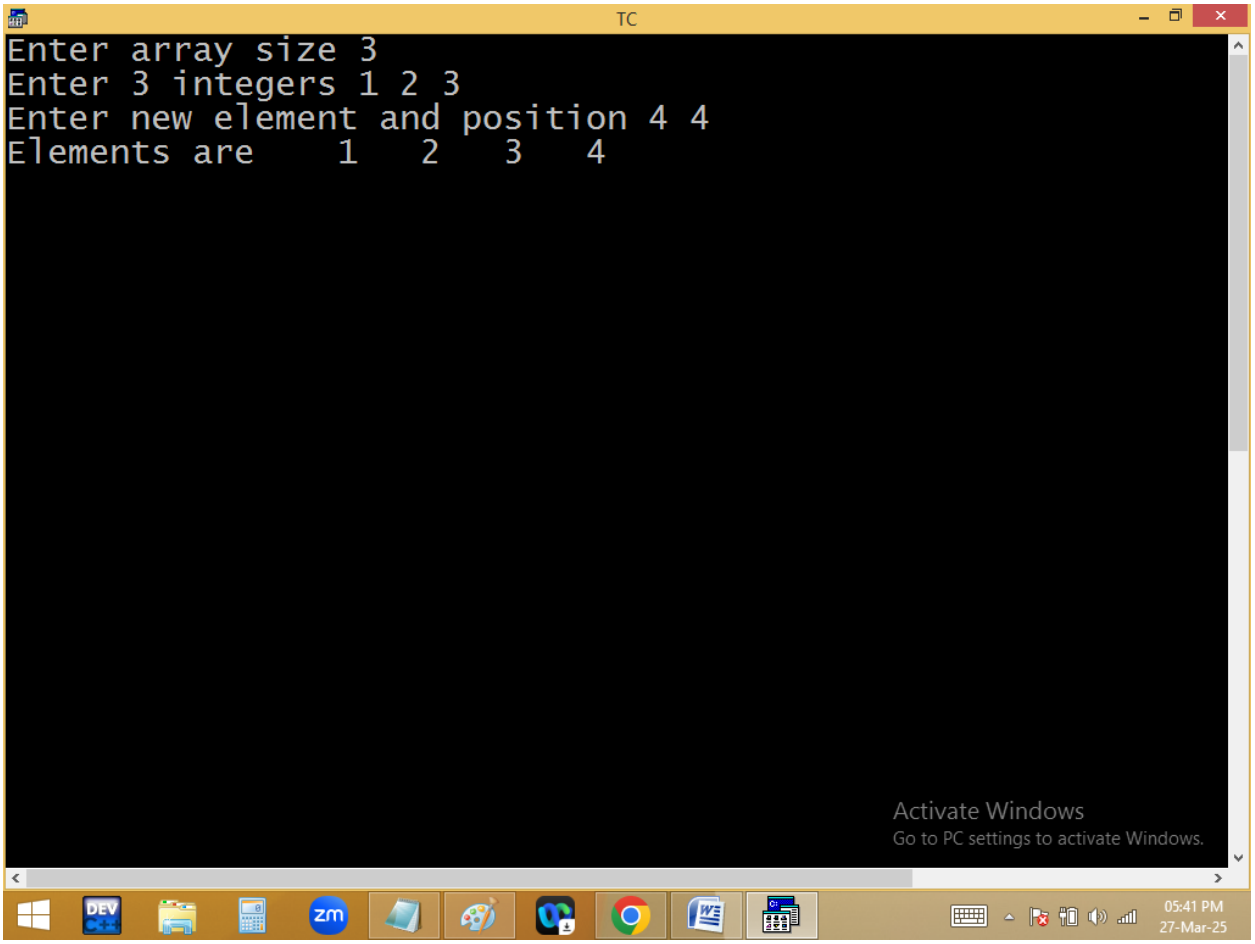
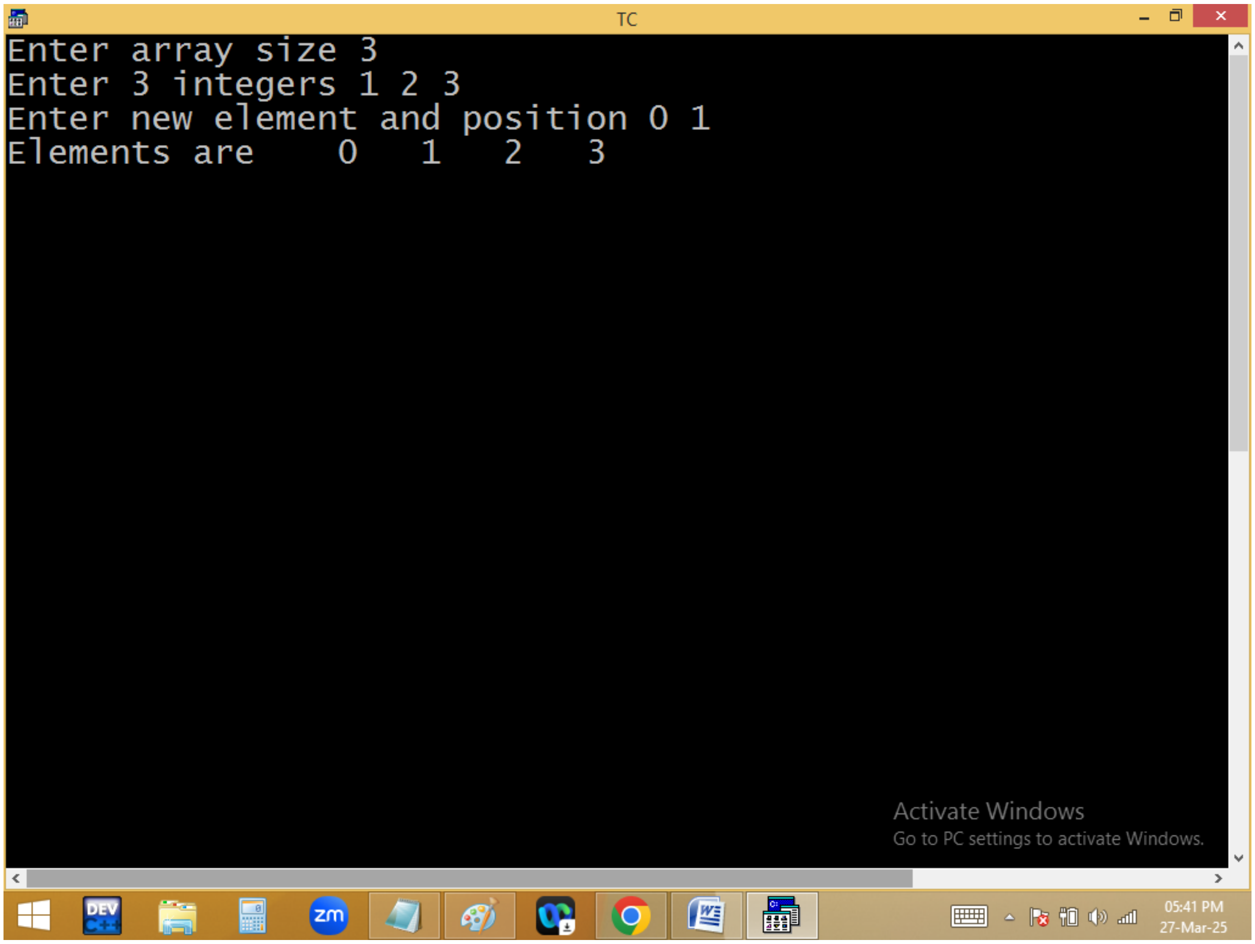
27-Mar-25


TC
Enter array size 3
Enter 3 integers 2 0 5
Enter new element and position 2 3
Elements are 2 0 2 5_
Activate Windows
Go to PC settings to activate Windows.
05:41 PM
27-Mar-25


TC
Enter array size 3
Enter 3 integers 1 2 3
Enter new element and position 4 4
Elements are   1   2   3   4
Activate Windows
Go to PC settings to activate Windows.
05:41 PM
27-Mar-25


TC
Enter array size 3
Enter 3 integers 1 2 3
Enter new element and position 0 1
Elements are 0 1 2 3
Activate Windows
Go to PC settings to activate Windows.
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2>=3

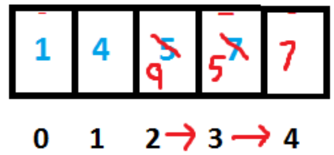
3>=3 a[3]=a[2]

i=4; 4>=3 ; 4-- a[4]=a[3];

```
for( i=n; i>=pos; i-- ) a[i] = a[i-1];
a[i]=ele;
```

2

a



$\frac{n}{4}$

$\frac{n}{3}$ $\frac{p}{9}$ 0

i=n p

3 >= 9

a[3]=9;

i=3;3>=0;i--

a[3]=a[2]

a[2]=a[1]

a[1]=a[0]

a[0]=a[-1]

Deleting particular element from array?